

- 3 A program stores records in the 2D array `HashTable`. Each record is stored at a specific index of the array that is calculated using a hashing algorithm with the record's key field.

The array has 100×10 elements. The hashing algorithm uses the key to generate an index between 0 and 99 (inclusive). If two key fields generate the same index, there is a collision. Any records that have a collision are stored in the next space in the same index.

For example: In this table two record keys generated the same hash value of 1. Four record keys generated the same hash value of 3.

Index	0	1	2	3	4	5	...	9
0	record							
1	record	record						
2								
3	record	record	record	record				
4								
...								
99								

The program uses Object-Oriented Programming (OOP).

- (a) The class `Record` stores data about the records:

Record	
Key : Integer	stores the integer key field for the data
Data : String	stores the string data
Constructor()	initialises <code>Key</code> and <code>Data</code> to its parameter values

The attributes `Key` and `Data` are public.

Write program code to declare the class `Record` and its constructor.

Use your programming language appropriate constructor.

Save your program as **Question3_N25**.

Copy and paste the program code into part 3(a) in the evidence document.

[2]